

SPORT CENTER MANAGEMENT SYSTEM

Software Requirement Document

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Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

Definition and Acronyms

Acronym	Definition
PWM	Psychology website
AWS	Amazon Web Services
BA	Business Analysis
BR	Business Rule
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
PM	Project Manager
SDD	Software Design Description
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
UAT	User Acceptance Test
UC	Use Case
API	Application Program Interface

I. Introduction

1. Product Background

Currently, many small and medium-sized sport centers in Vietnam still manage their daily operations manually or through disconnected tools such as Excel spreadsheets, paper forms, and Zalo/Facebook messages. Center staff record member information, class schedules, and payments by hand, while coaches keep track of student attendance and training progress in personal notebooks. This manual approach creates several problems: member and class data is often duplicated or lost, receptionists cannot quickly check whether a member's package is still valid, coaches have no structured way to record training results, and center managers have no real-time overview of revenue or class registration rates. In addition, members have no self-service channel to register for classes, check their training history, or communicate with their coach, so most requests must go through the front desk by phone or in person. As the number of members grows, this manual process becomes time-consuming, error-prone, and unable to scale, which motivates the need for a dedicated software system to manage the center's.

2. Existing Systems

2.1 System name1

MLC Sports Center is a sport training center in Ho Chi Minh City, belonging to the EQuest education ecosystem, offering 14–15 sports courses (swimming, basketball, volleyball, football, badminton, tennis, taekwondo, golf, chess, ballet, etc.) for students and members. Its website is mainly an informational/marketing site: it introduces the facility, coaches, and course list, and provides a simple "Tư vấn ghi danh" (enrollment inquiry) form where a visitor selects a sport and leaves their contact information for staff to call back.

Actors: Guest/parent (visitor), Center staff (offline, handling inquiries by phone).

Features: Course catalog by sport, coach profiles, facility introduction, and a lead-capture registration form.

Pros: Clear presentation of courses and facilities; good marketing content; easy for visitors to leave a request.

Cons: No member login/account system, no online class schedule or booking, no online payment, no way for an existing member to check their own class schedule, attendance, or training progress. All actual registration, scheduling, and payment must still be done manually/offline after the initial inquiry is received. This shows a real gap that a dedicated management system (like SCMS) could fill: for MLC-style centers, there is no self-service portal for members and no back-office tool for staff to manage classes, coaches, and payments digitally.

2.2 System name2

California Fitness & Yoga (Cali) is one of the largest fitness and yoga chains in Vietnam, operating more than 25 clubs nationwide. Beyond its marketing website, Cali provides a member mobile application called "Cali Life", along with website features such as "Weekly Schedule", "Find Class", and separate membership pricing pages (Standard, VVIP).

Actors: Member, Coach/Personal Trainer, Receptionist/staff at each club, System Admin (backend).

Features:

- Members can view the weekly class schedule and search for classes by club, time, or type (Yoga, Dance, Group X, Kickfit, etc.).

- Members can book classes and manage their profile through the Cali Life app.
- Membership packages are sold in tiers (Standard, VVIP) with different prices and benefits, displayed on dedicated pages.
- The company also promotes personal-trainer programs (health coaching, corrective exercise, RXPilates) as add-on services.

Pros: Large-scale, multi-branch system with a real member app for self-service class booking; clear separation between membership tiers; good UX for browsing schedules and services.

Cons: Designed for a large commercial chain (25+ clubs), so it is heavy and costly to replicate for a single, small sport center; class booking and account management are mainly member-facing, with limited public visibility into how staff manage coach assignment, attendance tracking, or reporting; no visible AI-based training recommendation feature.

3. Business Opportunity

Looking at the two existing systems, we can see a clear gap in the market. California Fitness & Yoga has a mobile app for booking classes, but it is built for a large chain with 25+ clubs, so it is too expensive and too complex for a single, small sport center to use. On the other hand, MLC Sports Center only has a marketing website with a simple contact form; there is no online system at all, so staff still have to manage registration, scheduling, and payment manually.

This shows that small and medium-sized sport centers need a system that is in between these two extremes: simple and affordable like MLC's needs, but still digital and convenient like Cali's app. Without such a system, centers still face common problems: members cannot check their class schedule or training history online, receptionists have to manually check membership status, coaches have no easy way to record attendance or training results, and managers cannot see revenue or membership reports in real time. The Sport Center Management System (SCMS) is built to solve these problems by giving a single center all the tools it needs in one affordable platform, plus extra AI features (like workout suggestions) that neither existing system currently has.

4. Software Product Vision

For sport centers that are still managing members and classes manually or with basic tools, the Sport Center Management System (SCMS) is a web and mobile application that lets Members register, book classes, and track their training progress online, while giving Center Managers, Coaches, and Receptionists one place to manage memberships, class schedules, attendance, and payments. Unlike MLC's website, which needs manual follow-up for every request, and unlike Cali's app, which is designed for a large multi-branch chain, SCMS is built to be simple and affordable for a single sport center. With SCMS, members no longer need to call or visit the center just to check their schedule, and staff can manage daily operations faster and with fewer errors.

5. Major Features

User & Membership Management

- FE-01: Register, authenticate, and manage profiles for Guest, Learner, Coach, Receptionist, and Center Manager accounts.
- FE-02: Register, renew, upgrade, or cancel membership packages, either by the Learner online or by the Receptionist on the Learner's behalf at the front desk.
- FE-03: Manage user accounts and system role permissions (add, edit, lock/unlock accounts) by the Center Manager.

Class Booking & Schedule Management

- FE-04: View class schedules, and register, cancel, or switch class registrations, either by the Learner online or with Receptionist assistance.
- FE-05: Manage facility data (classes, subjects, rooms, activity schedules) and assign Coaches to specific classes, by the Center Manager.
- FE-06: View personalized schedule information for each role (Learner's classes, Coach's teaching schedule, Receptionist's lookup).

Payment & Report Management

- FE-07: Record payments (cash or e-payment) for membership packages or services, and issue invoices, by the Receptionist.
- FE-08: View dashboards and reports on member count, class registration status, and revenue over a specified time period, by the Center Manager.

Training & Attendance Management

- FE-09: Record attendance for each class session (by Coach or Receptionist, including self check-in).
- FE-10: Record and track a Member's training results after each session, and allow the Coach to evaluate the Learner's progress with comments.
- FE-11: Create and manage training plans and learning materials for individual Members or entire classes.

AI Workout Recommendation

- FE-12: Provide AI-generated workout suggestions for a Coach, based on a Learner's goals, fitness level, and training history.

AI Assistant

- FE-1: Provide an AI chatbot that answers Learner and Coach questions about class schedules, exercises, or center services automatically.

6. Limitations and Exclusions

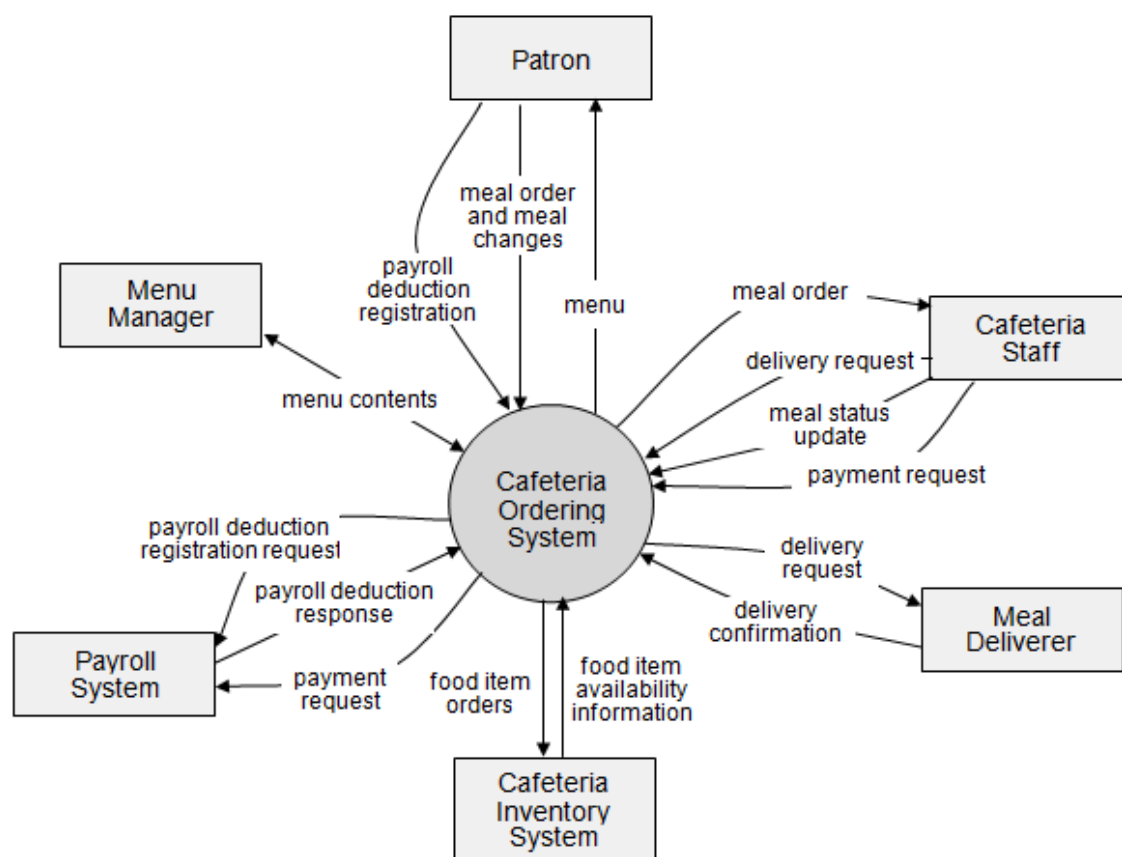
- LI-1: An early version of SCMS supports only a single sport center facility, operating in the Asia/Ho_Chi_Minh time zone with VND as the default currency; multi-branch or multi-location management is not included.
- LI-2: Each user account is restricted to exactly one primary role (Center Manager, Coach, Receptionist, or Learner); an account cannot hold multiple roles at the same time.
- LI-3: SCMS does not include hardware integration with turnstiles, access-control gates, or biometric check-in devices; attendance is recorded manually through the software interface by Coaches or Receptionists.
- LI-4: Online payment gateway integration (e.g., VNPay, MoMo) may be limited to a basic connection; full reconciliation with external accounting systems is out of scope.
- LI-5: The AI features (workout recommendation and AI chatbot) are optional, lower-priority features that may not be fully completed in the first release, and their outputs may require human review by a Coach before being sent to a Member.
- LI-6: SCMS does not manage retail sales or inventory of sport equipment/products; its scope is limited to member, class, coaching, attendance, and payment management.

II. Overall Description

1. Product Overview

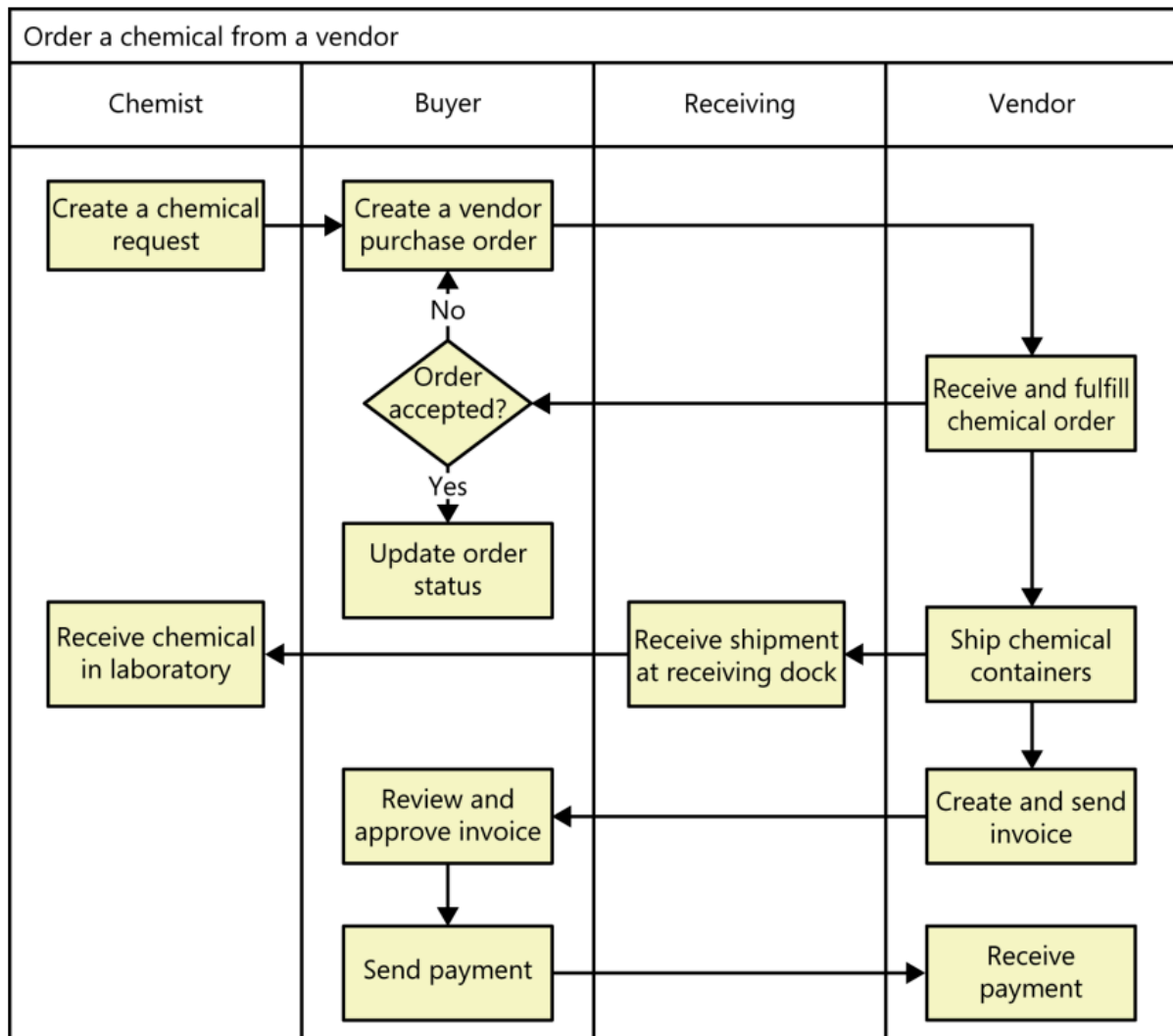
The Sport Center Management System (SCMS) is intended for an indoor sport center of the "Arena3" type: a mid-sized single-facility sport center located in a large city (reference model: Ho Chi Minh City), consisting of 8 badminton courts, 1 basketball court (28×15 m), 1 volleyball court, a reception area, 2 locker rooms, an equipment storage room, and a coach corner. The center operates from 06:00 to 22:00 every day of the week. In its first year of operation, the center targets a capacity of 400–800 active members with 150–250 visits per day, and is staffed by 1 Center Manager, 8–12 Coaches (full-time and part-time), and 2–3 Receptionists working in rotating shifts.

Before digitalization, the center's operations are assumed to be handled manually: court schedules are kept on Excel sheets and shared through Zalo groups, payments are recorded by hand in a notebook, coaches are sometimes double-booked for the same class or court, management has no visibility into peak-hour capacity, and members cannot check the status of their own membership package. SCMS is designed to replace this fragmented way of working with a single source of truth for people (accounts), membership packages, class/court schedules, and payments — reducing scheduling conflicts, manual data entry, and communication overhead between the front desk, coaches, and learners.



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2. Business Process



#	Process Step	Description
1	Create a chemical request	<<Process Step description>>
2	Create a vendor purchase order	
3	...	

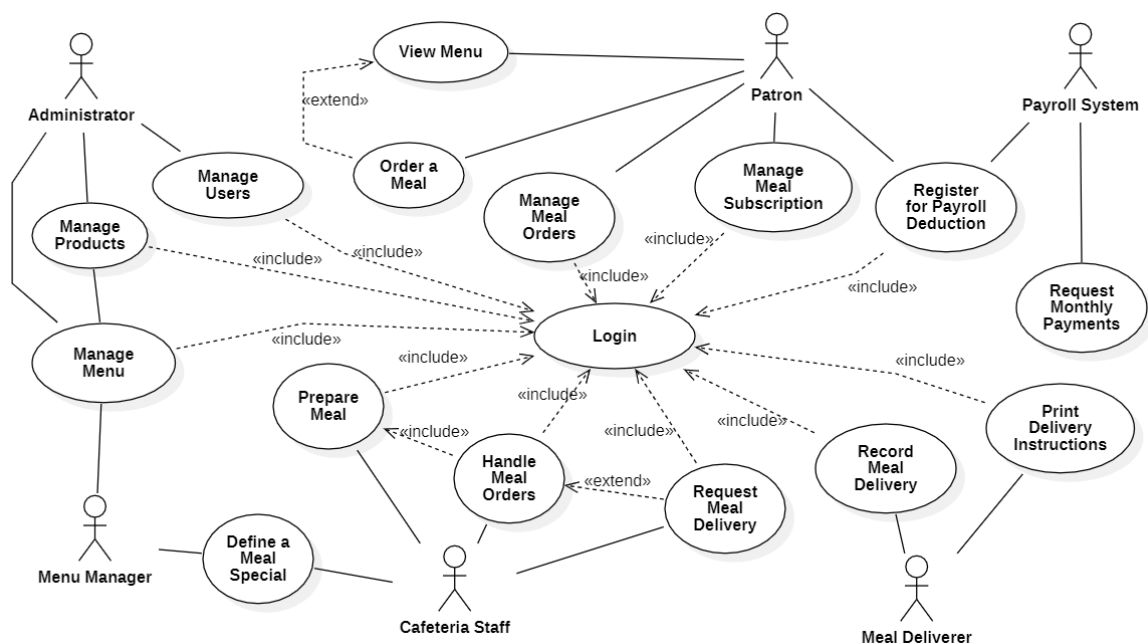
III. User Requirements

1. Actors

#	Actor	Description
1	Learner	A person who uses the sports center's services. Learners can manage their profile, subscriptions, classes, schedules, court bookings, and receive system notifications.
2	Coach	A trainer responsible for assigned classes. Coaches can view learner's information, prepare lesson plans, record attendance, evaluate learners, and optionally use AI-generated exercise recommendations.
3	Receptionist	A staff member who operates at the reception desk. Receptionists can register new learners, renew subscriptions, record facility check-ins, support class operations, process payments, and handle walk-in court bookings.
4	Center Manager	The person responsible for managing and administering the entire sports center. The Manager can manage staff, classes, courts, packages, permissions, revenue reports, and system audit logs.

2. Use Cases

2.1 Diagram



2.2 Descriptions

ID	Use Case	Actors	Use Case Description
UC-01	Register Account	Guest	Guests create a new account using email or phone number to access member features.
UC-02	Authenticate	Guest, Learner, Coach, Receptionist,	Users log into and out of the system, including via OAuth/OIDC.

		Center Manager, System Admin	
UC-03	Update Profile	Learner, Coach, Receptionist, Center Manager, System Admin	Users edit their personal information (name, avatar, phone number, training goals).
UC-04	Register Sport Center Visit	Guest, Receptionist	Guest registers for a visit to the Sport Center, either online or with the receptionist's help at the front desk.
UC-05	Manage Membership	Guest, Learner, Receptionist	Learner registers, renews, or upgrades a membership package; receptionists can do this on their behalf.
UC-06	Cancel Membership	Learner	Learners request to cancel their current membership; the system processes it per center policy.
UC-07	Manage Class Registration	Learner, Receptionist	Learner views classes, registers, cancels, or switches classes; receptionists can assist with these actions.
UC-08	Message Coach	Learner, Coach	Learner and coach exchange messages; learners can also view the coach's basic profile.
UC-09	Submit Coach Feedback	Learner	Learners submit feedback about a coach's teaching after a course or session.
UC-10	View and Moderate Feedback	Coach, Center Manager	The coach views feedback about them; the center manager reviews all feedback and can remove inappropriate content.
UC-11	Record Attendance	Coach, Receptionist	Coach checks in learners during sessions (including self check-in); receptionist checks in learners at the front desk.
UC-12	Track Workout Result	Coach, Learner	The coach records a learner's results per session; learner views their training history and progress.
UC-13	Evaluate Learner	Coach, Learner	The coach evaluates a learner's progress and gives advice; learner views past evaluations.
UC-14	Ask AI Chatbot	Learner, Coach, AI System	The user asks the AI System questions about schedules, exercises, or services and gets an automated answer.
UC-15	Get AI Workout Recommendation	Coach, AI System	The coach requests AI-generated exercise suggestions based on a learner's goals, level, and history.
UC-16	Send Notification	Learner, Coach, Notification Service	The system notifies learners of schedule changes or expirations; coaches can also send notifications or assignments.
UC-17	View Schedule	Learner, Coach, Receptionist	Each role views relevant schedule info: learner's classes, coach's teaching schedule, or receptionist's lookups.
UC-18	Manage Training Content	Coach	Coach creates, edits, or deletes training plans and learning materials for a learner or class.
UC-19	Search Learner Information	Receptionist, Coach	Basic learner information is searched and viewed to support front-desk or coaching tasks.
UC-20	Register New Learner	Receptionist	The receptionist creates a new learner profile at the front desk for a walk-in customer.

UC-21	Process Payment	Receptionist, Guest, Payment Gateway	A payment is recorded (cash or e-payment) and an invoice is issued after the transaction.
UC-22	Record Complaint	Receptionist	Receptionist logs support requests or complaints from learners.
UC-23	Manage User Account	Center Manager	Center manager adds, edits, locks, or removes accounts of learners, coaches, and staff.
UC-24	Manage Facility Data	Center Manager	The center manager manages classes, subjects, rooms, and activity schedules.
UC-25	Assign Coach	Center Manager	The center manager assigns a coach to a specific class.
UC-26	Manage Membership Plan	Center Manager	Center manager sets up membership packages, fees, and validity periods.
UC-27	Manage Staff Permission	Center Manager	Center manager configures system access permissions for each staff role.
UC-28	View Dashboard	Center Manager	Center manager views reports on member count, class registrations, and revenue over time.
UC-29	Monitor System Log	Center Manager, System Admin	Important system actions are logged and reviewed for auditing purposes.
UC-30	Manage System Role	System Admin	Admin creates/edits system roles and permissions, and locks/unlocks accounts.
UC-31	Configure System	System Admin	Admin manages system settings and integrations with external services (payment, AI, OAuth).

ID and Name:	UC-01 Register Account		
Created By:	Group 1	Date Created:	13/06/2026
Primary Actor:	Guest	Secondary Actors:	MySQL Database
Description:	A guest registers a new account in the Sport Center Management System by providing their personal information such as email or phone number. The system validates the data and creates a new member account to allow the guest to access member features.		
Trigger:	A guest wants to create an account to use the system's services.		
Preconditions:	PRE-1. The registration page is accessible. PRE-2. The database connection is available. PRE-3. The guest does not already have an account with the same email or phone number.		
Postconditions:	POST-1. A new account is created and stored in the system. POST-2. The guest receives a confirmation message or email. POST-3. The guest can now log into the system with the registered credentials.		
Normal Flow:	1.0 Register Account 1. The guest opens the registration page. 2. The system displays the registration form. 3. The guest enters personal information including full name, email or phone number, and password. (see 1.1)		

	4. The guest agrees to the Terms and Conditions and the personal data protection policy (Decree 13/2023). (see 1.0.E1) 5. The guest submits the registration form. (see 1.0.E2, 1.0.E3) 6. The system validates the entered information. 7. The system creates a new account with the role of Learner. 8. The system sends a confirmation message to the guest. 9. The guest is redirected to the login page or automatically logged in.
Alternative Flows:	1.1 Register via OAuth/OIDC 1. The guest clicks the "Sign in with Google" button. 2. The system redirects the guest to the OAuth provider. 3. The guest authenticates with the OAuth provider. 4. The system receives the guest's profile data from the provider. 5. Return to step 6 of normal flow.
Exceptions:	1.0.E1 Guest does not agree to Terms and Conditions 1. The system does not allow the guest to continue registration. 2. The system displays a message requiring agreement to the Terms and Conditions. 3. Return to step 3 of normal flow. 1.0.E2 Email or phone number already exists 1. The system detects a duplicate email or phone number. 2. The system displays an error message informing the guest that the account already exists. 3. Return to step 3 of normal flow. 1.0.E3 Invalid input data 1. The system detects invalid or missing required fields. 2. The system displays a validation error message. 3. Return to step 3 of normal flow.
Priority:	High
Frequency of Use:	Used frequently when new guests want to join the sport center. Usage is expected to be high in the first weeks after the system launches.
Business Rules:	BR-01, BR-02, BR-07, BR-08
Other Information:	The guest must provide consent to the Terms and Conditions before the account is created. Accounts with the same email or phone number are not allowed.
Assumptions:	The guest has a valid email address or phone number. The guest has internet access to complete the registration.

ID and Name:	UC-02 Authenticate		
Created By:	Group 1	Date Created:	13/06/2026
Primary Actor:	Guest, Learner, Coach, Receptionist, Center Manager	Secondary Actors:	MySQL Database, OAuth/OIDC Provider

Description:	A user logs into or out of the Sport Center Management System using their registered credentials or via OAuth/OIDC. The system verifies the identity of the user and grants access based on their assigned role.
Trigger:	A user wants to access the system's features and opens the login page.
Preconditions:	PRE-1. The login page is accessible. PRE-2. The database connection is available. PRE-3. The user already has a registered account in the system.
Postconditions:	POST-1. The user is successfully authenticated and redirected to the dashboard corresponding to their role. POST-2. If logout is performed, the user's session is terminated and they are redirected to the login page.
Normal Flow:	2.0 Authenticate (Login) 1. The user opens the login page. 2. The system displays the login form. 3. The user enters their email or phone number and password. (see 2.1) 4. The user submits the login form. (see 2.0.E1, 2.0.E2, 2.0.E3) 5. The system validates the credentials against the database. 6. The system identifies the user's role. 7. The system creates a session and redirects the user to their role-specific dashboard.
Alternative Flows:	2.1 Login via OAuth/OIDC 1. The user clicks "Sign in with Google" or another OAuth provider. 2. The system redirects the user to the OAuth provider login page. 3. The user authenticates with the provider. 4. The system receives the authentication token and identifies the user account. 5. Return to step 6 of normal flow. 2.2 Logout 1. The user clicks the "Logout" button. 2. The system terminates the user's session. 3. The system redirects the user to the login page.
Exceptions:	2.0.E1 Incorrect credentials 1. The system cannot verify the email/phone or password. 2. The system displays an error message stating that the credentials are incorrect. 3. The system increments the failed login counter for the account. 4. Return to step 3 of normal flow. 2.0.E2 Account is locked 1. The system detects that the account has been locked (after 5 failed attempts within 15 minutes, or locked by a Manager). 2. The system displays a message informing the user that the account is locked. 3. The use case terminates. 2.0.E3 Database is unavailable 1. The system cannot connect to the database. 2. The system displays a system error message. 3. The use case terminates.

Priority:	High
Frequency of Use:	Very frequently used. Every user must authenticate before accessing any feature. Expected to be used hundreds of times per day.
Business Rules:	BR-02, BR-04, BR-05
Other Information:	After 5 failed login attempts within 15 minutes, the account is automatically locked for 15 minutes. If a staff account is locked by a Manager, it cannot be unlocked using OTP.
Assumptions:	The user remembers their registered credentials. The network connection is stable during the login process.

ID and Name:	UC-03 Update Profile		
Created By:	Group 1	Date Created:	13/06/2026
Primary Actor:	Learner, Coach, Receptionist, Center Manager	Secondary Actors:	MySQL Database
Description:	A logged-in user updates their personal information in the system, such as full name, avatar, phone number, or training goals. The system saves the updated information and reflects changes immediately.		
Trigger:	A user wants to update their personal information in the system.		
Preconditions:	PRE-1. The user is logged into the system. PRE-2. The profile edit page is accessible. PRE-3. The database connection is available.		
Postconditions:	POST-1. The user's profile information is updated and saved in the database. POST-2. The system displays a success message to the user.		
Normal Flow:	3.0 Update Profile 1. The user navigates to the profile settings page. 2. The system displays the current profile information. 3. The user edits the desired fields such as full name, phone number, avatar, or training goals. 4. The user submits the changes. (see 3.0.E1, 3.0.E2) 5. The system validates the updated information. 6. The system saves the updated profile to the database. 7. The system displays a success message.		
Alternative Flows:	3.1 User cancels the update 1. The user clicks the "Cancel" button before submitting. 2. The system discards all changes. 3. The system returns the user to the previous page.		
Exceptions:	3.0.E1 Invalid input data 1. The system detects invalid or missing required fields (e.g., empty name, invalid phone format). 2. The system displays a validation error message. 3. Return to step 3 of normal flow.		

	3.0.E2 Phone number already used by another account 1. The system detects that the new phone number belongs to another account. 2. The system displays an error message. 3. Return to step 3 of normal flow.
Priority:	Medium
Frequency of Use:	Occasionally used. Users typically update their profile when they first register or when their personal information changes.
Business Rules:	BR-01, BR-08
Other Information:	Users can only edit their own profile. A Center Manager may edit other users' profiles through the account management function (UC-23).
Assumptions:	The user is aware of their correct personal information. The new phone number provided is not already registered in the system.

ID and Name:	UC-05 Manage Membership		
Created By:	Group 1	Date Created:	13/06/2026
Primary Actor:	Guest, Learner, Receptionist	Secondary Actors:	MySQL Database, Payment Gateway
Description:	A learner registers, renews, or upgrades a membership package in the Sport Center Management System. The receptionist can also perform these actions on behalf of a learner at the front desk. The system records the subscription and updates the learner's membership status accordingly.		
Trigger:	A learner wants to register for a new membership package, renew an existing one, or upgrade to a higher-tier package.		
Preconditions:	PRE-1. The user is logged into the system (or the receptionist is logged in at the front desk). PRE-2. The membership management page is accessible. PRE-3. The database connection is available. PRE-4. Membership packages are configured and available in the system.		
Postconditions:	POST-1. The learner's membership subscription is created, renewed, or upgraded in the system. POST-2. The membership status is updated to Active (or Pending if payment is incomplete). POST-3. A payment record is created in the system.		
Normal Flow:	5.0 Register Membership Package 1. The learner (or receptionist on the learner's behalf) navigates to the membership management page. 2. The system displays the list of available membership packages. 3. The learner selects a desired membership package. (see 5.1, 5.2) 4. The system displays the package details including price, duration, and scope. 5. The learner confirms the selection and proceeds to payment. (see 5.0.E1) 6. The system processes the payment.		

	7. The system activates the membership and updates the learner's subscription status. 8. The system displays a confirmation message and generates an invoice.
Alternative Flows:	5.1 Renew existing membership 1. The learner selects "Renew" on their current membership package. 2. The system calculates the new end date based on BR-10 or BR-11. 3. Return to step 5 of normal flow. 5.2 Upgrade membership package 1. The learner selects a higher-tier package to upgrade. 2. The system checks the learner's current subscription status. 3. The system applies the new package and updates the subscription scope. 4. Return to step 5 of normal flow.
Exceptions:	5.0.E1 Payment fails 1. The payment gateway returns a failure response. 2. The system displays an error message to the user. 3. The membership is not activated. 4. Return to step 5 of normal flow.
Priority:	High
Frequency of Use:	Frequently used when new learners join or existing members renew their packages. Expected to be used regularly throughout the month.
Business Rules:	BR-08, BR-10, BR-11, BR-12, BR-13, BR-16, BR-19, BR-47, BR-49, BR-50, BR-52, BR-54
Other Information:	A learner can only have one active subscription per sport scope at a time. If the learner is under 16 years old, the guardian's information must be provided before activation.
Assumptions:	The learner has a valid payment method. Membership packages have been configured by the Center Manager in advance.

ID and Name:	UC-06 Cancel Membership		
Created By:	Group 1	Date Created:	13/06/2026
Primary Actor:	Learner	Secondary Actors:	MySQL Database
Description:	A learner requests to cancel their current membership package. The system processes the cancellation according to the center's cancellation and refund policies and updates the learner's subscription status.		
Trigger:	A learner wants to cancel their current active membership package.		
Preconditions:	PRE-1. The learner is logged into the system. PRE-2. The learner has at least one active membership package. PRE-3. The database connection is available.		
Postconditions:	POST-1. The learner's membership status is updated to Canceled. POST-2. If eligible, a refund is processed according to the center's refund policy. POST-3. The system records the cancellation with a reason.		

Normal Flow:	6.0 Cancel Membership 1. The learner navigates to the membership management page. 2. The system displays the learner's current active membership packages. 3. The learner selects the membership package they want to cancel. 4. The system displays the cancellation policy and potential refund information. 5. The learner confirms the cancellation request. (see 6.1) 6. The system checks the eligibility for a refund based on BR-54. 7. The system updates the membership status to Canceled. 8. The system records the cancellation and displays a confirmation message.
Alternative Flows:	6.1 Learner cancels within 24 hours of purchase 1. The system detects that the cancellation request is within 24 hours of purchase and no check-in, class, or court booking has been used. 2. The system applies a 100% refund as per BR-54. 3. The system processes the refund and updates the subscription status. 4. Return to step 8 of normal flow.
Exceptions:	6.0.E1 No active membership found 1. The system detects that the learner does not have an active membership package. 2. The system displays an informational message. 3. The use case terminates.
Priority:	Medium
Frequency of Use:	Occasionally used. Learners do not frequently cancel memberships; this use case is expected to be used infrequently.
Business Rules:	BR-14, BR-47, BR-48, BR-54, BR-73
Other Information:	Membership records are not hard-deleted from the system. They are only marked as canceled. Refunds of VND 1,000,000 or more require Manager approval.
Assumptions:	The learner understands the cancellation policy before confirming. The refund (if any) will be processed through the original payment method.

3. Business Rules

ID	Rule Definition
BR-01	A phone number can be associated with only one account. A learner ID in the format (A3-YYYY-nnn) is assigned only once and cannot be reused, even if the profile is deactivated.
BR-02	The password must contain at least 8 characters, including both letters and numbers. Passwords must be stored using Argon2id/bcrypt hashing, never in plaintext, and must not be returned through the API.
BR-03	V1.x operates with only one facility, using the Asia/Ho_Chi_Minh time zone and VND as the default currency.
BR-04	After 5 failed password attempts within 15 minutes, the account is locked for 15 minutes. If a staff account is locked by a Manager, it cannot be unlocked using an OTP.
BR-05	In v1.0, each account must have exactly one primary role: Manager, Coach, Receptionist, or Learner. A single account cannot hold multiple roles simultaneously.

BR-06	The system must not deactivate or downgrade the last remaining Active Manager account.
BR-07	For learners under 16 years old (configurable), the system requires the guardian's full name and phone number before activating a membership package.
BR-08	During registration (via app or at the reception desk), the user must provide consent to the Terms and Conditions and Decree 13/2023 on personal data protection. If consent is not provided, the system must not create the profile.
BR-09	When a Coach/Receptionist account is created by a Manager, the staff member must change the temporary password upon their first login before accessing system functions.
BR-10	$\text{new_end} = \text{old_end} + \text{duration of the additional package}$. Two packages with the same <code>sport_scope</code> must not run concurrently.
BR-11	If $\text{old_end} < \text{today}$, then $\text{new_end} = \text{today} + \text{duration of the package}$. The period between the previous expiration date and the new start date is not counted.
BR-12	A subscription with status Pending/Expired/Frozen does not grant permission to enroll in classes or rent courts, and does not increase quota. A debt approved by a Manager is an exception subject to the limit defined in BR-44.
BR-13	The system sends expiration notifications 7 days, 3 days and the expiration date. The notification milestones are configurable.
BR-14	Only a Manager can freeze a package due to injury or temporary leave. Frozen services are treated as expired. The <code>end_on</code> date remains unchanged unless the Manager explicitly chooses to add the freezing period.
BR-15	The maximum freezing period is 30 days per calendar year per learner by default. This limit is configurable. Any request exceeding the limit must be rejected.
BR-16	A single-subject package allows the learner to enroll only in classes and rent courts for the specified subject. An all-access package can be used for all three subjects.
BR-17	Court rental hours included in a package expire when the package expires. By default, unused hours do not carry over to the next period. A Manager may enable the carry-over option for individual plans.
BR-18	For session-based packages, each confirmed enrollment or Present attendance (if configured) consumes 1 session credit. When the remaining quota reaches 0, no additional enrollment is allowed.
BR-19	A learner can have only one Active subscription for the same <code>sport_scope</code> . Purchasing an additional package must extend the existing subscription according to BR-10/BR-11, rather than creating concurrent subscriptions.
BR-20	If a learner cancels an enrollment at least 4 hours before the session starts, the slot is released and the next learner on the waitlist is notified. If canceled less than 4 hours before the session, the slot is retained, the enrollment is marked canceled-late, and the session credit is not refunded (for session-based packages).
BR-21	The system must prevent overlapping Sessions (and Session + CourtBooking occupancy) on the same <code>court_id</code> . Two classes cannot be scheduled with the same coach at overlapping times.
BR-22	Badminton : 8-16 learners, table tennis: 10-16 learners, volleyball: 10-18 learners . A Manager may adjust the capacity for each class, but the maximum capacity must not exceed the facility's capacity.
BR-23	A coach may be assigned only to a class that matches a subject for which the coach is qualified. Any required additional coach qualifications must also be satisfied.

BR-24	A learner cannot enroll in two classes whose sessions overlap in time. The system must reject the enrollment attempt.
BR-25	The waitlist is managed according to First-In, First-Out (FIFO) based on registration time. When a slot becomes available, the first learner in the queue is notified. If the learner does not claim the slot within the configured period (2 hours by default), the slot is offered to the next learner.
BR-26	When a Manager cancels a session, the system must notify the Coach and Learners within 2 minutes. For session-based packages, one session credit is restored. Time-based packages are not refunded in cash.
BR-27	A coach may be replaced only by another coach who teaches the same subject. The system must notify the Learner of the change. The original coach loses the corresponding session on their teaching schedule.
BR-28	A session may be rescheduled at least 12 hours before its original start time (configurable). Rescheduling within a shorter period requires the Manager to provide a reason. The new time slot must comply with BR-21.
BR-29	Each person may have a maximum of one trial session per subject for their lifetime. A trial session does not require a subscription and does not consume package quota. Trial sessions are not available for competitive/team classes.
BR-30	Court rentals are sold in 60-minute slots, with bookings starting on the hour between 06:00 and 21:00. 30-minute slots are not available in Version 1.0. Slot duration may be configurable in future versions.
BR-31	A soft hold reserves a court slot for 5 minutes while payment is being processed. When the hold expires, the slot is automatically released. A user may have only one active hold at a time (BR-39).
BR-32	Court bookings can be made up to 7 days in advance, with a maximum of 2 slots per day per user (configurable). Walk-in bookings at the reception desk are not subject to the 7-day advance limit and may be made only for the current or next available session.
BR-33	If a court booking is canceled at least 2 hours before the start time, the user receives a 100% refund or quota restoration. If canceled less than 2 hours before the start time, no refund is provided. If the user does not check in by 10 minutes after the scheduled start time, the booking is marked as NoShow and is not refundable.
BR-34	A learner with an Active package covering the relevant subject (or an all-access package) receives the configured percentage discount for that plan. Walk-in customers pay the listed price. Each booking consumes 1 rental-hour quota, regardless of the court's tier or quality level.
BR-35	The system must not allow bookings with a start_at outside the 06:00–22:00 operating hours or on days when the facility is closed, including holidays and full-day maintenance periods.
BR-36	Maintenance schedules and events must block the corresponding court and time slot in the same manner as a confirmed booking. The system must not allow overlapping bookings.
BR-37	When BR-01 is converted to a temporary competitive court, all BR-01 and BC-01 bookings/classes that overlap with the conversion period must be blocked until the conversion is deactivated.

BR-38	A grace period of 10 minutes after end_at is allowed. If the next slot is available, the receptionist may sell an additional slot on-site. If the next slot has already been booked, the customer must vacate the court.
BR-39	When a user creates a new hold, the system must automatically cancel the user's previous unpaid hold.
BR-40	For walk-in customers, the system must require the customer's full name and phone number on the receipt. A system account is not required.
BR-41	Court check-in is allowed from 15 minutes before to 10 minutes after start_at. Check-in attempts outside this window are rejected, unless a Receptionist overrides the restriction, in which case the override must be recorded in the audit log.
BR-42	If a learner attempts to book a court that overlaps with their own class session, the system must issue a warning. The booking is allowed only if the member explicitly confirms it. The system must not automatically cancel the class enrollment.
BR-43	The price, discount percentage, and VAT are fixed at the time the booking is Confirmed. Subsequent changes to the pricing configuration must not be applied retroactively to existing confirmed bookings.
BR-44	If a learner has overdue unreturned equipment, they cannot create new court bookings through the app until the outstanding equipment has been returned and recorded as returned by staff, or the Manager clears the equipment debt.
BR-45	Version 1.0 does not support recurring court bookings. Each court slot must be booked as a separate booking. Recurring or weekly reservations may be considered in Version 2.0.
BR-46	The court rental price includes lighting. No additional lighting fee is charged in Version 1.0.
BR-47	Payment receipts must not be deleted or have their amounts modified. Cancellations and refunds must be recorded as corresponding reversal transactions with negative amounts, together with a reason.
BR-48	Refunds of VND 1,000,000 or more require Manager approval. Refunds below the configured threshold may be processed by the Receptionist according to BR-33/BR-47.
BR-49	The VAT rate applicable at the time of payment must be recorded on the invoice. Changes to the VAT rate must not modify previously issued invoices.
BR-50	All monetary amounts must be rounded to the nearest VND 1,000, with half-up rounding.
BR-51	If a learner's outstanding debt exceeds VND 500,000 (configurable), the system automatically blocks the services specified in BR-12 until the outstanding debt is fully paid.
BR-52	If enabled, a payment of at least 50% of the package fee activates the subscription as Active, while the remaining amount is recorded as outstanding debt. By default, full payment is required before activation.
BR-53	A discount code cannot be combined with the package's percentage discount, unless the promotion is explicitly configured with stackable = true.
BR-54	A 100% refund is provided if the cancellation request is submitted within 24 hours of purchase, provided that there has been no center check-in, class attendance, or confirmed court booking.
BR-55	Payments made using quota must not be included in cash or bank-transfer revenue. Financial reports must record quota usage separately as "quota consumption."

BR-56	At the end of each shift, total cash collected must equal the sum of all receipts with method = cash for that shift. The system must flag any discrepancy but must not automatically lock the shift in Version 1.0.
BR-57	Attendance records are locked 2 hours after the session ends. Any modification after the lock requires Manager authorization and a reason.
BR-58	Facility check-in does not automatically mark class attendance, and class attendance does not replace facility check-in. These two records must be managed independently.
BR-59	Only the Coach assigned to the member's class, Receptionist (for emergencies), and Manager may view health_notes.
BR-60	Completing or failing to complete homework does not deduct class sessions and does not generate any additional payment.
BR-61	A learner is marked Present if they check in during the first 10 minutes of the session. After that, the member is marked Late if the Coach records the late arrival. If the learner does not attend the session, they are marked Absent.
BR-62	After 3 consecutive unexcused absences, the system must notify the Manager and Coach. The system must not automatically cancel the learner's enrollment.
BR-63	A Coach may record attendance only for sessions where they are assigned as the primary or assistant Coach.
BR-64	AI-generated training plans must remain in Draft status. A plan can be converted to TrainingPlan status only when the Coach explicitly approves or saves it.
BR-65	Questions regarding schedules, packages, or pricing must be answered using the database retrieval tool for the currently authenticated user. If no relevant data is available, the AI must state that it does not have the information and must not guess or fabricate an answer.
BR-66	For injuries or illnesses, the AI must advise the user to consult a Coach or seek appropriate medical care. The AI must not diagnose medical conditions or prescribe therapeutic exercises.
BR-67	When the F5/F6 feature flags are disabled, the system must not invoke the LLM, must hide the AI-related UI, and AI API requests must return HTTP 403 (Forbidden).
BR-68	Prompts must not contain national ID numbers, phone numbers, other students' data, or payment card information. Prompt logs must mask or redact personally identifiable information (PII).
BR-69	Schedule changes/cancellations, confirmed court bookings, package expiration, and waitlist promotions must generate in-app notifications. SMS/email notifications are optional and depend on system configuration; they must not replace the required in-app notification.
BR-70	Audit logs must be append-only. They must not be edited or deleted through the user interface and must be retained for at least 12 months.
BR-71	Disabling F4, F5, or F6 must not affect the operation of F1, F2, F3, or F7. Feature flags must be managed by the Manager.
BR-72	When the reception desk loses network connectivity, the system must not allow the creation of local bookings or payment receipts. This restriction prevents double booking and inconsistent payment records.

BR-73	Users, sold packages, and payment receipts must not be hard-deleted. They may only be disabled/deactivated, marked as no longer available for sale, or handled through the corresponding reversal transaction/document.
BR-74	Changes to pricing, quotas, cancellation windows, or default class sizes apply only to new transactions or classes. Existing bookings and enrollments retain their original snapshots and must not be modified retroactively.
BR-75	The app must not allow users to create court bookings with a start_at earlier than the current time. Walk-in bookings may be created for an ongoing session only if at least 20 minutes remain.
BR-76	Classes with status = draft must not be displayed in the learner-facing class catalog. Draft classes are visible only to the Manager and assigned Coaches.
BR-77	Every API endpoint must enforce role-based access control (RBAC) on the server side. Hiding or disabling UI buttons does not constitute authorization.
BR-78	Each payment transaction must have exactly one unique payment document/receipt number in the format(PAY-YYYYMMDD-nnnn) . The document number must not be reused.

IV. Functional Requirements

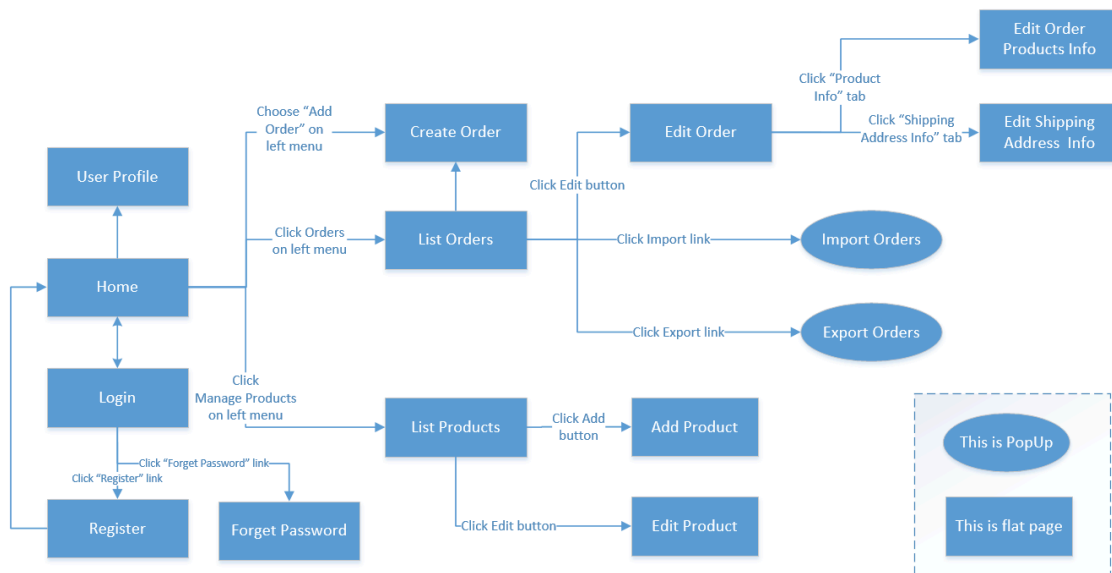
1. System Functional Overview

[Provide functionality overview of software system: screen flow, screen descriptions, system user roles, screen authorization, non-screen functions, ERD]

1.1 Screen Flow

[This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below]

<<Sample:



>>

1.2 Screen Descriptions

[Provide the descriptions for the screens in the Screens Flow above]

<<Sample:

#	Screen	Feature	Description
FR-01	Create Order	Order Meals	<<Screen Brief description>>
FR-02	Change Order	Order Meals	
FR-03	..		

>>

1.3 Screen Authorization

[Provide the system roles authorization to the system features (down to screens, and event to the screen activities if applicable) in the table form as below – replace Role1, Role2,... with the specific system user role names]

<<Sample:

Screen	Role1	Role2	Role3	Role4	RoleX
<<Screen Name1>>	X			X	X
<<Screen Activity>>				X	X
<<Screen Name2>>	X			X	
Query All Data	X				
Query Own Data				X	
Query Managed Data				X	

Add New Data				X	X
Update All Data					X
Update Own Data					X
Update Managed Data					X
Delete Data					
...					

In which:

- Role1: <<role1 description>>
- Role2: <<role2 description>>
- ...

>>

1.4 Non-Screen Functions

[Provide the descriptions for the non-screen system functions, i.e batch/cron job, service, API, etc.]

<<Sample:

#	System Function	Feature	Description
FR-11	<<Function Name1>>	<<Feature Name>>	<<Function Name1 Description>>
FR-12	...		

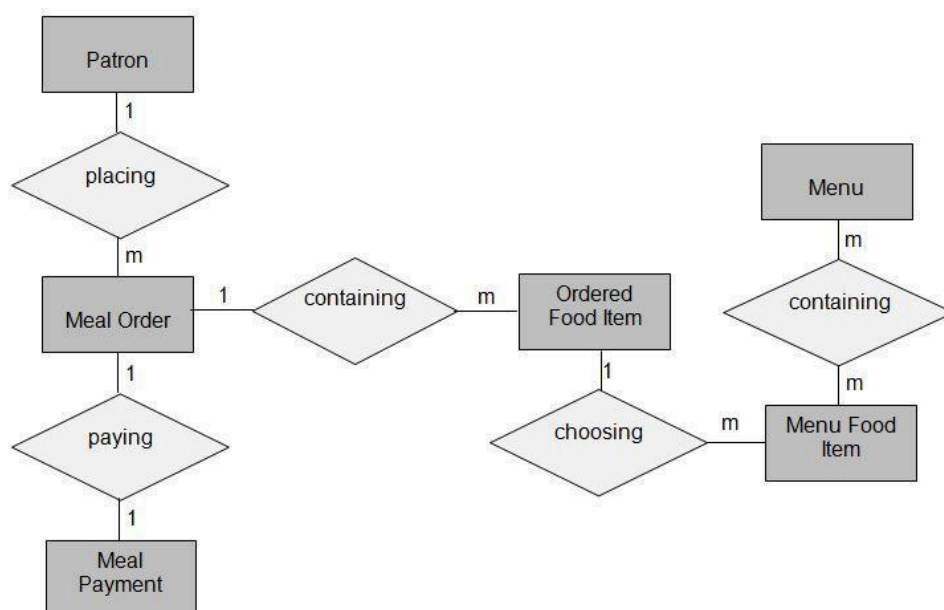
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2. Data Requirements

2.1 Logical Data Model

[A data model is a visual representation of the data objects and collections the system will process and the relationships between them. Include a data model for the business operations being addressed by the system, or a logical representation for the data that the system itself will manipulate. Data models are most commonly created as an entity-relationship diagram.]

<<Sample:



>>

2.2 Data Dictionary

[The data dictionary defines the composition of data structures and the meaning, data type, length, format, and allowed values for the data elements that make up those structures. In many cases, you're better off storing the data dictionary as a separate artifact, rather than embedding it in the middle of an SRS. That also increases its reusability potential in other projects.]

<<Sample:

Data Element	Description	Composition or Data Type	Length	Values
delivery instruction	where and to whom a meal is to be delivered, if it isn't being picked up in the cafeteria	patron name +patron phone number +meal date +delivery location +delivery time window		
delivery location	building and room to which an ordered meal is to be delivered	alphanumeric	50	hyphens and commas permitted
delivery time window	beginning time of a 15-minute range on the meal date during which an ordered meal is to be delivered;	time	hh:mm	local time; hh = 0-23 inclusive; mm = 00, 15, 30, or 45
employee ID	company ID number of the employee who placed a meal order	integer	6	
food item description	ext description of a food item on a menu	alphabetic	100	
food item price	pre-tax cost of a single unit of a menu food item	numeric, dollars and cents	dd.cc	
meal date	the date the meal is to be delivered or picked up	date, MM/DD/YYYY	10	default = current date if the current time is before the order cutoff time, else the next day; cannot be prior to current date
meal order	details about a meal a Patron ordered	meal order number + order date + meal date + 1:m{ordered food item} + delivery instruction + meal order status		
meal order number	unique ID that COS assigns to each accepted meal order	integer	7	initial value is 1

meal order status	status of a meal order that a Patron initiated	alphabetic	16	incomplete, accepted, prepared, pending delivery, delivered, canceled
meal payment	information about a payment COS accepted for a meal	payment amount + payment method + transaction number		
menu	list of food items available for purchase on a specific date	menu date + 1:m{menu food item}		
menu date	the date for which a specific menu is available	date, MM/DD/YYYY	10	
menu food item	description of a menu item	food item description + food item price		
order cutoff time	the time of day before which all meal orders for that date must be placed	time, HH:MM	5	
order date	the date on which a patron placed a meal order	date, MM/DD/YYYY	10	
ordered food item	one menu food item that a Patron requested as part of a meal order	menu food item + quantity ordered		
patron	a Process Impact employee who is authorized to order a meal	patron name + employee ID + patron phone number + patron location + patron email		
patron email	email address of the employee who placed a meal order	alphanumeric	50	
patron location	building and room numbers of the employee who placed a meal order	alphanumeric	50	hyphens and commas permitted
patron name	name of the employee who placed a meal order	alphabetic	30	
patron phone number	telephone number of the employee who placed a meal order	AAA-EEE-NNNN xXXXX for area code (A), exchange (E), number (N), and extension (X)	18	
payment amount	total price of an order in dollars and cents, calculated per BR-12	numeric, dollars and cents	dddd.c c	
payment method	how the Patron is paying for a meal he ordered	alphabetic	16	payroll deduction,

				cash, credit card, debit card
quantity ordered	the number of units of each food item that the Patron is ordering in a single meal order	integer	4	default = 1; maximum = quantity presently in inventory
transaction number	unique sequence number that COS assigns to each payment transaction	integer	12	

>>

2.3 Reports

[If your application will generate any reports, identify them here and describe their characteristics. If a report must conform to a specific predefined layout you can specify that here as a constraint, perhaps with an example. Otherwise, focus on the logical descriptions of the report content, sort sequence, totaling levels, and so forth, deferring the detailed report layout to the design stage.]

<<Sample:

#	Report Name	Description
RPT-01	Ordered Meal History	Patron wants to see a list of all meals that he had previously ordered from the Process Impact cafeteria or local restaurants over a specified time period up to six months prior to the current date, so he can reorder a particular meal he liked.
RPT-02	...	

Report ID:	RPT-01
Report Title:	Ordered Meal History
Report Purpose:	Patron wants to see a list of all meals that he had previously ordered from the Process Impact cafeteria or local restaurants over a specified time period up to six months prior to the current date, so he can reorder a particular meal he liked.
Priority:	Medium
Report Users:	Patrons
Data Sources:	Database of previously placed meal orders
Frequency and Disposition:	Report is generated on demand by a Patron. Data in the report is static. Report is displayed on user's web browser screen on a computer, tablet, or smartphone. It can be printed if the display device permits printing.
Latency:	Complete report must be displayed to Patron within 3 seconds after it is requested.
Visual Layout:	Landscape mode
Header and Footer:	Report header shall contain the report title, Patron's name, and date range specified. If printed, report footer shall show the page number.
Report Body:	Fields shown and column headings: • Order Number • Meal Date • Ordered From ("Cafeteria" or restaurant name) • Items ordered (list all items in the meal order, their quantity, and their prices) • Total Food Price

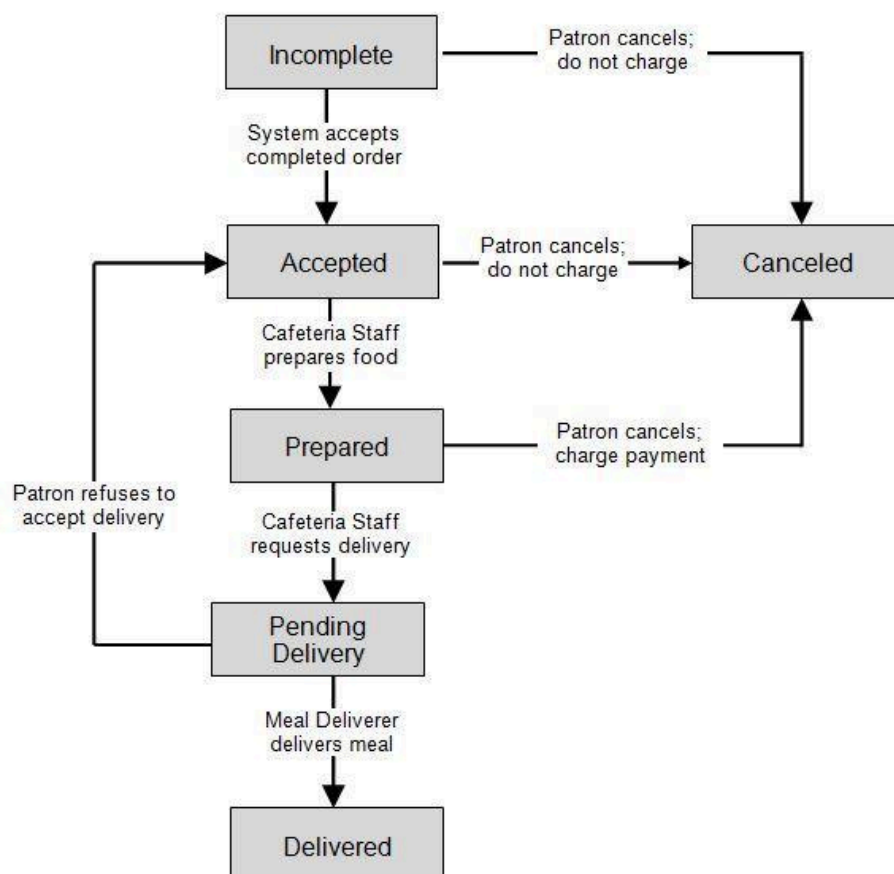
	• Tax • Delivery Charge • Total Price (sum of food item prices, tax, and delivery charge) Selection Criteria: date range specified by Patron, inclusive of end points Sort Criteria: reverse chronological order
End-of-Report Indicator:	None
Interactivity:	Patron can drill down to see ingredients and nutritional information for each item in the order
Security Access Restrictions:	A Patron may retrieve only his own meal order history

>>

2.4 State transition diagram

[This part shows state transition of main entities in the system]

<<Sample: a state-transition diagram that shows the possible meal order statuses and the allowed changes in status.



>>

3. <<Feature Name 1>>

3.1 <<Function Name 1>>

[A function can be a screen or a non-screen function (listed in the part 5.1 above). In this part, you need to provide the details on the related function, focus on mentioning below information]

- *Function trigger: how this function is triggered (navigation path, a timing frequency, etc.*
- *Function description: actors/roles, purpose, interface, data processing, etc.*
- *Screen layout: mockup prototype of the screen, sample below is for Manage Products screen*

LOGO HERE

Data Administration
Manage Users
Manage Products
Manage Menu
Order Handling
Orders List
Payments

Screen Header

Manage Products

All Categories
Type the product code or name to search
Search

Add new

Id	Product	Code	Category	Price	Action
25	Coca Cola	DR03	Drink	20.000 VND	Edit Delete
31	Chicken	FD156	Food	135.000 VND	Edit Delete
38	Meatball Soup	FD134	Food	249.000 VND	Edit Delete
39	Pasta & Meat ball	FD146	Food	199.000 VND	Edit Delete
45	Coca Cola	DR03	Drink	20.000 VND	Edit Delete
51	Chicken	FD156	Food	135.000 VND	Edit Delete
58	Meatball Soup	FD134	Food	249.000 VND	Edit Delete
59	Pasta & Meat ball	FD146	Food	199.000 VND	Edit Delete
62	Meatball Soup	FD134	Food	249.000 VND	Edit Delete
63	Pasta & Meat ball	FD146	Food	199.000 VND	Edit Delete

1 2 3 4 5

- *Function Details: provide explanation for the data, validation, business logics, functionalities (for both normal cases and abnormal cases), etc. of the function so that the reader can image how it work.*

]

3.2 <<Function Name 2>>

...

4. <<Feature Name 2>>

...

V. Non-Functional Requirements

1. External Interface Requirements

1.1 User Interfaces

- UI-1: The Arena 3 system screen displays shall conform to the Progressive Web App (PWA) and responsive design standards for mobile and desktop viewports.
- UI-2: The system shall provide an interactive court matrix scheduler to visualize real-time slot availability, hold states, and maintenance periods.
- UI-3: The web application shall permit full navigation and court slot selection using mouse, keyboard, or mobile multi-touch interactions.
- UI-4: The system shall display non-blocking error notifications with actionable resolution instructions whenever a domain exception (such as slot conflict or expired hold) occurs.

1.2 Software Interfaces

- SI-1: Neon Serverless PostgreSQL Database: The system shall connect to Neon PostgreSQL via TLS-encrypted connection pooling to execute ACID-compliant transactional queries.
- SI-2: Embedded PGLite WASM: The system shall support an in-process WebAssembly PostgreSQL engine as an offline and development fallback database..
- SI-3: Payment Gateway (VietQR / Banking): The system shall interface with payment providers via HTTPS REST APIs to generate dynamic QR codes and process payment reconciliations..
- SI-4: Notification Outbox Service: The system shall deliver SMS OTP verifications and waitlist notifications through a transactional outbox queue with idempotency keys..

1.3 Hardware Interfaces

HI-1: Server Configuration

Application Server (Vercel / Cloud Container)

CPU : Virtual CPU (vCPU) 2 Core 2.50 GHz

Memory Space : 4 GB RAM

Storage Space : 50 GB SSD NVMe

Operating System : Linux Ubuntu Server 22.04 LTS

Software : Node.js 22.x LTS, TanStack Start Runtime

Database Server (Neon Serverless PostgreSQL)

CPU : Autoscaling Compute (0.25 to 2 CU)

Memory Space : 1 GB to 8 GB RAM (Dynamically Scaled)

Storage Space : 10 GB (Autoscaling Cloud Storage)

Operating System : Managed Linux Container

Software : PostgreSQL 16

HI-2: Client Configuration

PC Device (Manager / Receptionist Desktop)

CPU	: Intel Core i3 / AMD Ryzen 3 2.40 GHz or higher
Memory Space	: 4 GB RAM DDR4
Storage Space	: SSD 128 GB
Operating System	: Windows 10 / Windows 11 / macOS
Operator Display	: 19.5-inch to 24-inch widescreen, 16:9 format 1920 x 1080 pixel (Full HD recommended), 1366 x 768 pixel (Min)
POS Device (Front-Desk Counter)	
CPU	: Intel® Celeron® J1900 / Core i3 2.0 GHz Dual-Core
Memory Space	: 4 GB DDR3 / DDR4
Storage Space	: SSD 64 GB
Operating System	: Windows 10 IoT Enterprise / Windows POSReady
Operator Display	: 15" TFT color LCD capacitive touch screen 1024 x 768 pixel, 1366 x 768 pixel Option: Thermal Receipt Printer (USB/LAN, 80mm) Option: 1D/2D QR Barcode Scanner
Mobility Device (Member & Coach Smart Devices)	
CPU	: Octa-Core 1.6 GHz or Apple A-Series / M-Series
Memory Space	: 3 GB RAM minimum
Storage Space	: 32 GB minimum
Operating System	: Android 9.0+ / iOS 15.0+
Wifi Standard	: Wi-Fi 802.11 a/b/g/n/ac (2.4 GHz & 5 GHz), 4G/5G LTE
Operator Display	: 5.5 inches to 11.0 inches 390 x 844 Pixels (Min), Multi-Touch Screen
HI-3: Network	
LAN Network	: Speed ≥ 100 Mbps (Center Local Network)
WAN Network	: Speed ≥ 50 Mbps / 20 Users operate together

1.4 Communications Interfaces

- CI-1: The Arena 3 system shall send an SMS or in-app notification to the Member or Guest to confirm court reservation details, assigned court code, scheduled time, and payment status.
- CI-2: The system shall send an SMS or in-app notification to the Member to deliver one-time password (OTP) verification codes for authentication and sensitive account actions.
- CI-2: The system shall send an automated notification to enrolled Members on the class waitlist when an opening becomes available, specifying the acceptance deadline.
- CI-2: The system shall send an SMS or in-app notification to affected Members and Coaches to report any schedule modifications, court cancellations, or facility maintenance alerts.

2. Quality Attributes

2.1 Usability

- USE-1: A new Learner shall be able to successfully register an account and enroll in a class without any assistance on their first attempt.
- USE-2: The system shall display all error messages in plain language that clearly explains what went wrong and how the user can fix it.
- USE-3: The class schedule and membership status shall be visible to a Learner within 2 clicks or taps from the main dashboard.
- USE-4: The Receptionist shall be able to complete a walk-in learner registration and process a membership payment in under 5 minutes using the front-desk interface.
- USE-5: The system shall support both desktop and mobile browsers with a responsive layout that adjusts to screen sizes from 390px (mobile) to 1920px (desktop).

2.2 Performance

- PER-1: The system shall support a minimum of 400 registered users and handle up to 100 concurrent users during peak hours (06:00–09:00 and 17:00–20:00 local time), with an estimated average session duration of 10 minutes.
- PER-2: 95% of all page requests shall fully load within 3 seconds over a network connection of 10 Mbps or faster.
- PER-3: The system shall display booking confirmation or payment confirmation to the user within an average of 3 seconds and a maximum of 6 seconds after the user submits the request.
- PER-4: The court slot availability matrix shall refresh in real time (within 2 seconds) when any booking status changes, to prevent double-booking.
- PER-5: Attendance recording and class check-in operations shall complete within 2 seconds per learner entry to avoid delays during session start.

2.3 Security

- SEC-1: All data transmitted between the client and server shall be encrypted using HTTPS/TLS 1.2 or higher, including all payment and personally identifiable information (PII).
- SEC-2: All users must authenticate before accessing any system feature except the public registration page.
- SEC-3: Every API endpoint shall enforce role-based access control (RBAC) on the server side as specified in BR-77. Hiding UI elements alone does not constitute authorization.
- SEC-4: Passwords must be stored using hashing and must never be stored in plaintext or returned through any API response, as per BR-02.
- SEC-5: After 5 consecutive failed login attempts within 15 minutes, the account shall be automatically locked for 15 minutes, as per BR-04.
- SEC-6: AI prompts must not contain national ID numbers, phone numbers, payment card information, or any other PII. Prompt logs must mask or redact all PII before storage, as per BR-68.
- SEC-7: Audit logs shall be append-only, retained for at least 12 months, and must not be editable or deletable through any user interface, as per BR-70.

2.4 Safety

- SAF-1: The system shall display a clear warning to the Learner if a court booking they are attempting to create overlaps with their own enrolled class session, before allowing the action to proceed, as per BR-42.
- SAF-2: When a Manager cancels a class session, the system shall send notifications to all affected Coaches and Learners within 2 minutes to minimize disruption, as per BR-26.
- SAF-3: The AI chatbot shall not diagnose medical conditions or prescribe therapeutic exercises. For any injury or illness-related queries, the AI must advise the user to consult a Coach or seek appropriate medical care, as per BR-66.
- SAF-4: The system shall not allow the last remaining active Center Manager account to be deactivated or downgraded, ensuring the center always has at least one administrator, as per BR-06.

2.5 Availability

- AVL-1: The system shall be available at least 99% of the time during center operating hours (06:00–22:00, Asia/Ho_Chi_Minh time zone), excluding scheduled maintenance windows.
- AVL-2: Scheduled maintenance shall be performed outside of operating hours (22:00–06:00) and must be announced to staff at least 24 hours in advance.
- AVL-3: When the reception desk loses network connectivity, the system shall not allow the creation of local bookings or payment receipts to prevent double-booking and inconsistent records, as per BR-72.

2.6 Reliability

- REL-1: The system shall not lose more than 1 confirmed booking or payment transaction out of every 10,000 transactions due to software failures.
- REL-2: The mean time between failures (MTBF) for the core booking and payment modules shall be at least 30 days of continuous operation.
- REL-: In the event of a system failure, all database transactions shall follow ACID properties to ensure no partial or corrupted records are saved.

2.7 Design Constraints

- DES-1: Frontend framework: React 19 with TanStack Start (SSR/SSG).
- DES-2: Backend runtime: Node.js 22.x LTS.
- DES-3: Primary database: PostgreSQL 16 via Neon Serverless (TLS-encrypted connection pooling).
- DES-4: Offline/development fallback database: PGLite (WebAssembly in-process PostgreSQL).
- DES-5: Authentication: OAuth 2.0 / OIDC (Google provider), with SMS OTP fallback via Notification Outbox Service.
- DES-6: Payment integration: VietQR / Banking REST API (HTTPS).
- DES-7: Deployment platform: Vercel (serverless container).
- DES-8: Supported browsers: Chrome 110+, Firefox 110+, Safari 16+, Edge 110+.
- DES-9: Supported mobile OS: Android 9.0+ and iOS 15.0+.
- DES-10: All monetary values shall be stored and calculated in VND, rounded to the nearest VND 1,000 using half-up rounding, as per BR-50.

2.8 Scalability

- SCA-1: The system architecture shall support horizontal scaling of the application server layer to accommodate growth beyond 800 active members without requiring code changes.
- SCA-2: The database shall use Neon's autoscaling compute (0.25 to 2 CU) to dynamically adjust resources based on load, avoiding manual capacity planning in Version 1.0.
- SCA-3: The feature flag system (F4–F6) shall allow optional modules (Training Management, AI Workout Recommendation, AI Assistant) to be enabled or disabled independently without affecting core flows F1–F3, as per BR-71.